Zhilin (Hector) He

Pittsburgh, PA | (267) 210-8618 | hectorh@andrew.cmu.edu | LinkedIn | GitHub

Education Background

Carnegie Mellon University, Pittsburgh, PA

- ***M.S. in Electrical and Computer Engineering*** 01/2025 – Expected 12/2026

University of Southern California, Los Angeles, CA

- ***B.S. in Applied and Computational Mathematics (Transfered from Boston University)*** 01/2022 – 12/2023

Boston University, Boston, MA

- ***Double Major in Applied Mathematics and Computer Science*** 09/2019 – 12/2021

Publications / In Submission (*Equal Contributions, † Advising/Equal Advising)

- **Zhilin He**, Guoquan Qi, Changliu Liu†."*Manifold Generative Annealing for Contact Constraint-aware Impedance Control with Reinforcement Learning Prior.*" *In Submission ICLR 2027*. [Paper][Code][Website]
- **Zhilin He**, Xinyuan Wang, Changliu Liu†."*Model-Based Geometry-Aware Generative Optimization for Constrained Locomotion Planning.*" *In Submission RA-L 2026*. [Paper][Code][Website]
- [**CoRL Spotlight 2026**] **Zhilin He**, Gauri Joshi†."*FedGuide: Diffusion-Prior Alignment and Value Baseline Guidance for Heterogeneous Federated Reinforcement Learning.*" *Conference on Robot Learning (CoRL), 2026*. [Paper] [Code][Website]
- **Zhilin He***, Jianlin Dou*, Changliu Liu†."*Foundations of Safe Motion Planning for Autonomous Vehicles: From Ego Planning to Space-Time Coordination.*" *In Submission IET Books* [Code]
- [**IROS 2026**] **Zhilin He**, Bowei Li, Jianlin Dou, Yuner Zhang, Changliu Liu†."*Motion Planning with Model-Based Diffusion via Constraint Optimization and Adaptive Scheduling.*" *International Conference on Intelligent Robots and Systems (IROS), 2026*. [Paper][Code][Website]
- [**RSS 2026**] **Zhilin He**, Yorai Shaoul, Jiaoyang Li†."*Model-Based Diffusion Optimal Control for Multi-Robot Motion Planning.*" *Robotics: Science and Systems (RSS), 2026*. [Paper] [Code][Website]
- [**ACC 2024**] Shuangge Wang*, **Zhilin He***, Zihao Xu*, Cymra Haskell†, Bhaskar Krishnamachari†."*Optimal Control for Antivirus Routing in Epidemiological-Based Heterogeneous Computer Network Clusters.*" *American Control Conference (ACC), 2024*. [Paper] [Code]

Professional Experiences

Reli. — ***Full-time, Software Development Engineer***, Los Angeles, CA 01/2024 – 01/2025

- **Developed an adaptive reinforcement learning system** for query optimization in a **10 TB**-scale database, automatically tuning index–pagination strategies and reducing tail latency by **23%** across heterogeneous workloads.
- **Redesigned the CI/CD and deployment pipeline** using Docker Swarm and containerized microservices, enabling zero-downtime migrations and multi-node rollouts across staging and production environments, and future Kubernetes migration.
- **Transitioned the core bidding engine from rule-based heuristics to a contextual learning model**, improving bid-price prediction accuracy and increasing cost–performance KPIs by **12%** across campaigns.

Ping Cares — ***Intern, Data Engineer***, Chicago, IL 07/2023 – 09/2023

- Implemented **automated data pipelines** and **analytics scripts** in Python/SQL, reducing manual reporting costs by **60%**.
- Optimized **database schemas and caching strategies** for **10× faster data processing**, integrated with **ML-based analytics modules** for real-time insights.

Simens — ***Intern, Data Engineer***, Shanghai, China 04/2023 – 06/2023

- Built a **regression model for predicting** patients' liver fat percentage and assessing potential **lesion transformation risk**.
- Constructed a **custom neural network architecture** for the dataset and, by adjusting the loss function, achieved enhanced performance compared to the previous machine learning algorithm with a **30% reduction in MSE** (Mean Squared Error).

CNIC, Chinese Academy of Sciences — ***Intern & Lab, Deep Learning Engineer***, Beijing, China 05/2021 – 08/2022

- Utilized an **NLP approach** (Natural Language Processing) to **extract keywords** from text files and match them with given entities.
- Implemented the **YOLO framework** (Object Detection algorithm) for **document layout analysis**, resulting in the successful establishment of **entity relations** within the **graph database**.

Project Experiences

GenerativeDynamics [Code][Website] 12/2025 – Present

- Developed an **open-source robotics framework** unifying **generative optimization, robot learning, and feedback control** through modular task, solver, and execution interfaces, enabling interchangeable algorithms and reusable controllers.
- Integrated **my publicaiton works** into reproducible simulation workflows spanning **planar navigation, 7-DoF arm avoidance, legged locomotion, and contact-rich manipulation**, with configurable benchmarks, metrics, and replay tools.

TennisBuddy [Code] [Website] 09/2025 – 12/2025

- Advised by Prof Ziad Youssfi (CMU) and **designing a ROS2-based autonomous robot for tennis ball retrieval**, integrating SLAM, and Nav2 navigation through a custom bridge that enables adaptive switching between global planner and local controller.

EdgeCloud [Code] 07/2025 – Present

- Developing a unified orchestration layer that **transforms any Kubernetes cluster into an elastic pool of multi-cloud GPUs, CPUs**, and **edge nodes for distributed ML workloads**, featuring a **Rust-based scheduler and Knative serverless runtime** for dynamic balancing of inference, reinforcement learning, and robotics simulation across heterogeneous compute resources.

Machine Learning in Finance [Code] 07/2023 – 08/2023

- Assisted Prof Stanislav Minsker (USC) for hands-on statistic pair trading strategy exploration and optimized portfolio return.

Teaching Experiences

- 2026 Fall (CMU): Teaching assistant for 18-860 Optimization.
- 2023 Fall (USC): Teaching assistant for MATH 447 Mathematics of Machine Learning.

Review Service

- CoRL 2026, ICLR 2027, ICRA 2027

Skills

Machine Learning & Robotics: PyTorch, RLlib, JAX, ROS2, Nav2

Systems & Infrastructure: Docker, Kubernetes, Knative, Rust, Go, C/C++, Python